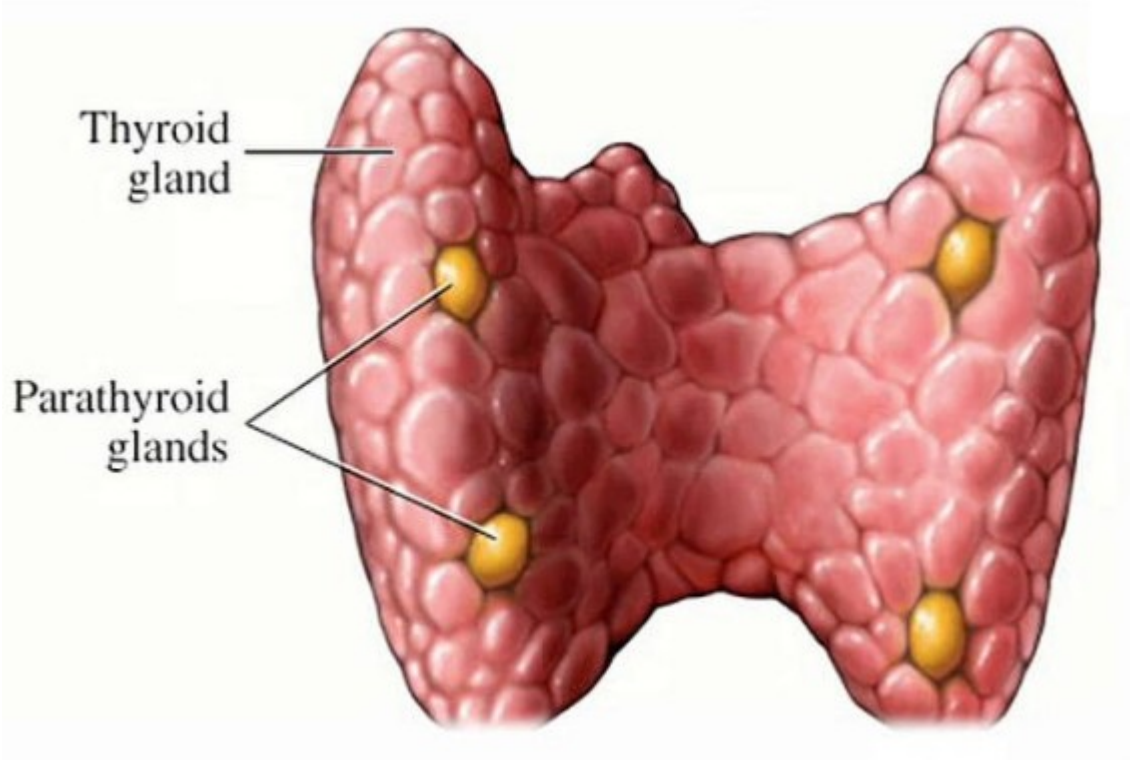
The Thyroid



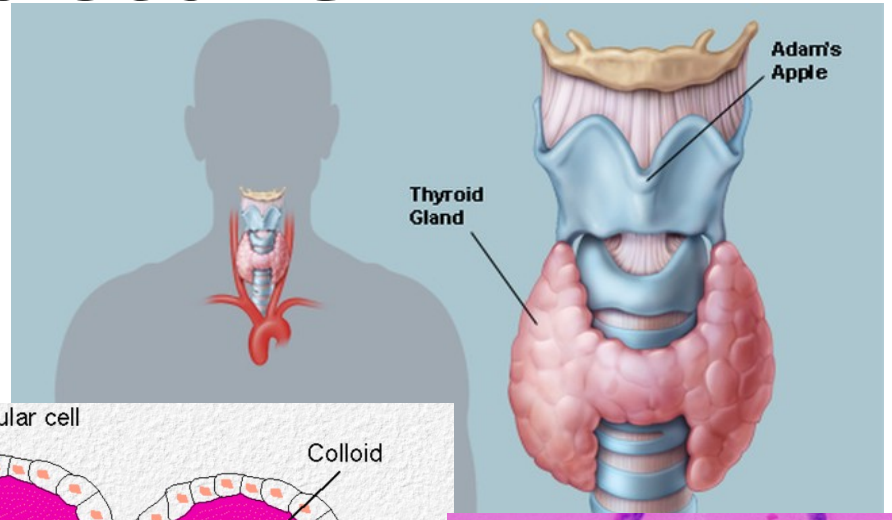
Hormones, Regulation and Disease

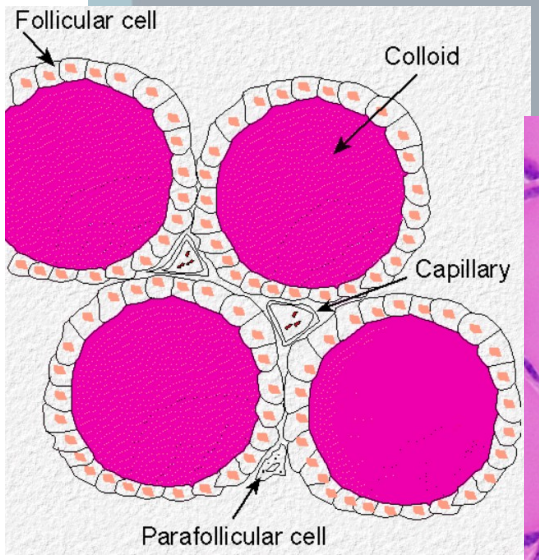
The Thyroid: Hormones and Regulation

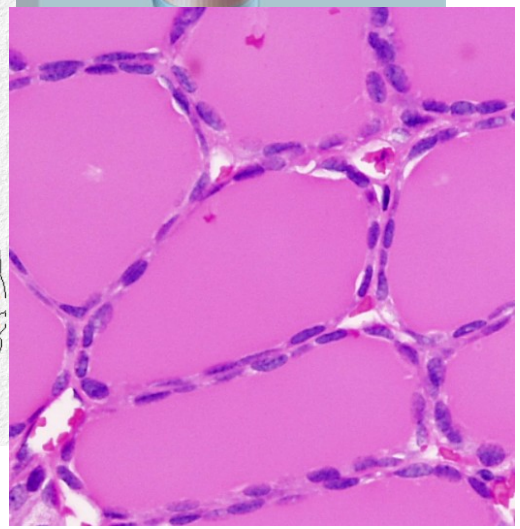


Thyroid location and structure

- Butterfly-shaped organ
- Located at base of neck
- Produces
  - Thyroxine (T4) – metabolic regulation
  - Triiodothyronine (T3) – metabolic regulation
  - Thyrocalcitonin – calcium homeostasis







Learning Aim: Describe the structure and location of the thyroid gland
Learning Aim: List hormones produced by the thyroid.

Thyroxine (T4)

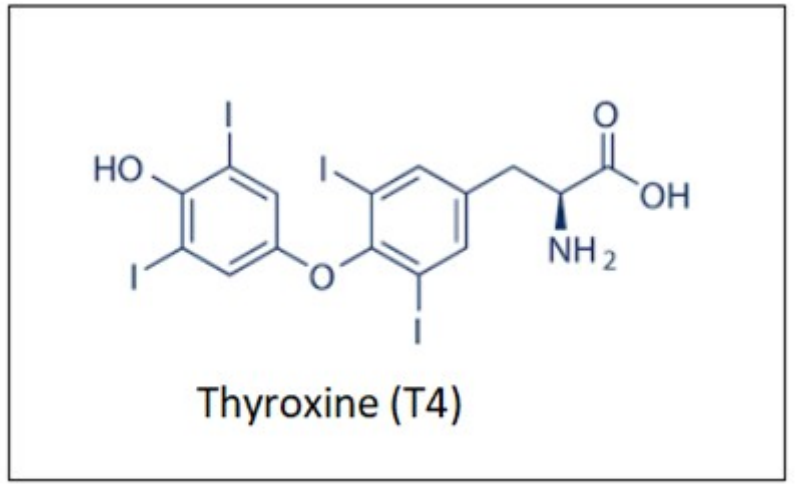
- Made by the Thyroid from Iodine consumed as part of the diet
- Regulates metabolic rate through wide range of effects:

  - Increases basal metabolic rate (BMR) and temperature by increasing catabolism of protein and carbohydrates

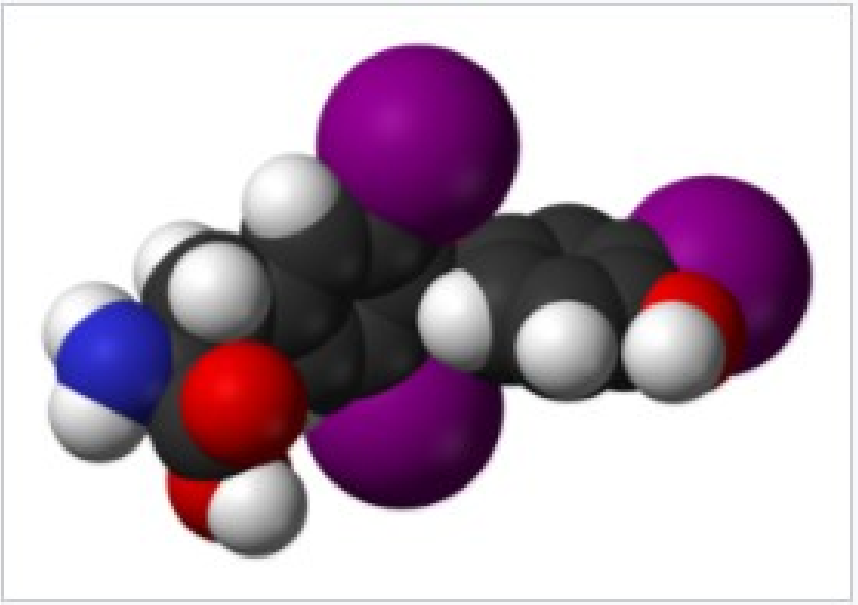
  - Increases heart rate, cardiac output, respiratory rate

  - Raises blood sugar
    - Converts glycogen to glucose
    - Increases absorption of sugar from intestine into blood

  - Assists with brain development in utero



Thyroxine (T4)



Learning Aim: Explain in detail how thyroxine is regulated and its effects on the body.

Regulation of Thyroxine production

Hypothalamus

Detects:

- Low levels of Thyroid hormone in blood
- Low metabolic rate
  (including low body temperature, low blood sugar and other parameters)

The hypothalamus produces **Thryotropin Releasing Factor (TRF)** *which travels through the blood vessels of the infundibulum to the Anterior Pituitary*

Anterior Pituitary

The TRF stimulates the glandular cells of the Anterior Pituitary to produce **Thyroid Stimulating Hormone (TSH)**

TSH *is released from the Anterior Pituitary and travels through the systemic circulation*

Thyroid

The TSH binds to receptors in the Thyroid, stimulating it to produce **Thyroxine**

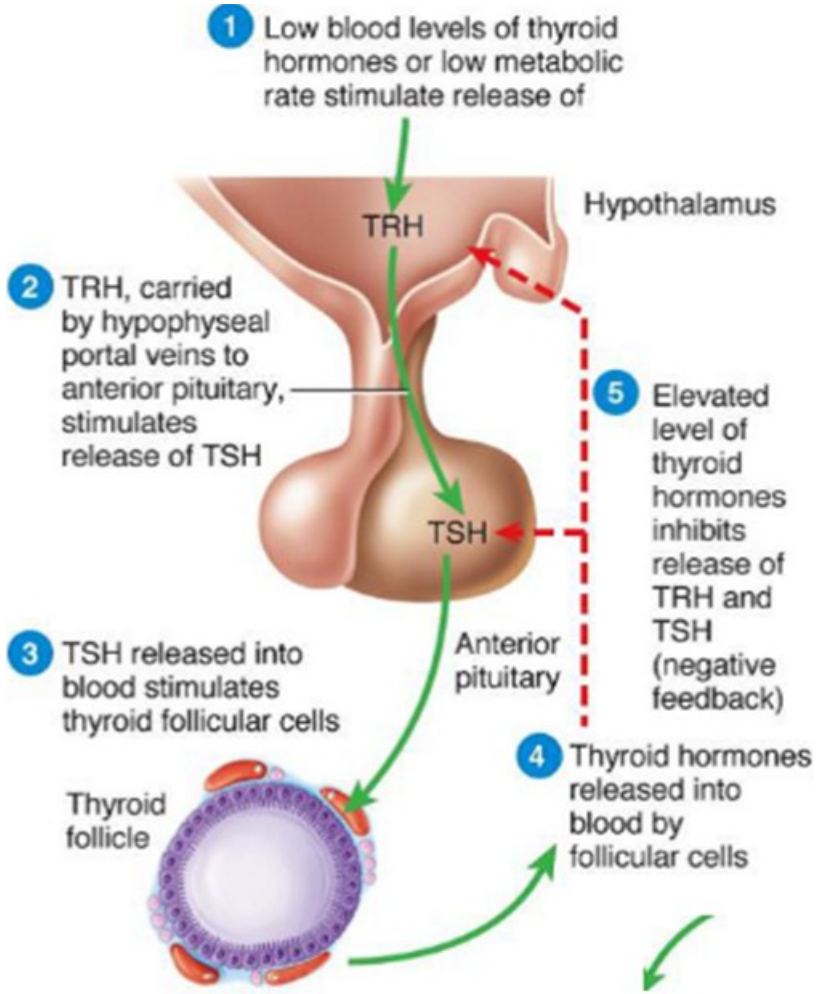
Thyroxine *is released from the Thyroid and travels through the systemic circulation*

Body Tissues

Thyroxine binds to receptors within most body tissues, which then respond by:

- Increasing metabolic rate and body temperature by increasing catabolism of protein and carbohydrates.
- Heart rate, blood pressure and respiratory rate are increased.
- Increases blood sugar by stimulating liver to convert glycogen to glucose and by stimulating intestines to absorb sugar.

Rising levels of Thyroxine inhibit further production of TRF and TSH, so that Thyroxine levels stabilise. This is **Negative Feedback**



Learning Aim: Explain in detail how thyroxine is regulated and its effects on the body.

Effects in Body

Increased Body Temperature
Increased BMR
Increased Blood Glucose
etc.

Thyroid hormone imbalance and disease

- Problems with Thyroid hormone production affect homeostasis, causing disease.

- ***Hyperthyroidism*** – Too much Thyroid hormone produced

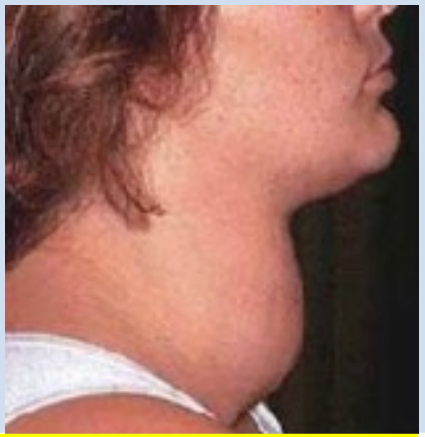


- ***Hypothyroidism*** – Too little Thyroid hormone produced

Hyperthyroidism – too much thyroid hormone

Common causes:	Symptoms	Diagnosis	Treatment
Graves' Disease – immune system produces antibodies which stimulate thyroid to produce hormones	Weight loss (or gain) Increased appetite Protruding eyeballs Fast heart rate Sleep disturbances Anxiety/nervousness Fatigue Muscle weakness Goitre	Blood test for TSH – lower than normal. Why? Testing for antibodies Radioactive iodine uptake by thyroid Ultrasound/CT scan/MRI	Radioactive iodine therapy • given as medication • taken up by thyroid destroys thyroid cells • Less thyroid hormone produced Anti-thyroid medication • Prevents thyroid from using iodine to make hormones Surgery
Thyroid nodules/adenoma – iodine deficiency causes overgrowth of thyroid tissue to try to produce more hormone, which also impacts on TSH production.			
Thyroiditis – inflammation of the thyroid causes overproduction of hormone			





Learning Aim: Discuss disorders related to over and undersecretion of thyroxine

Hypothyroidism – too little thyroid hormone

Common Causes	Symptoms	Diagnosis	Treatment
Hashimoto's Disease – autoimmune disease attacks thyroid tissue, destroying it so it cannot produce hormone. **Thyroid destroyed** by medication or surgery to correct hyperthyroidism **Not enough dietary iodine** – mostly a problem in developing nations – body can't produce hormone without iodine	Fatigue Weakness Weight gain Depression Cold intolerance Dry, rough skin Brittle dry hair Goitre 	Blood test – low levels of T4, high levels of TSH (why?) unless pituitary problem.	Thyroid hormone replacement medication: • *Levothyroxine* • Synthetic thyroid hormone tablets • Replace missing hormone 

Learning Aim: Discuss disorders related to over and undersecretion of thyroxine

The Thyroid and Thyrocalcitonin

- The thyroid gland also makes *Thyrocalcitonin* which is produced if Ca^{2+} levels in the blood are too high. **think back to Yr 11. Why would high blood Ca^{2+} be a problem?*

- Thyrocalcitonin reduces Ca^{2+} in the blood by:

  - ✓ Preventing Ca^{2+} from being reabsorbed by the kidney (so more is excreted in urine).
  - ✓ Promoting Ca^{2+} deposition in the bones (laying calcium down into the bone matrix).
  - ✓ Decreasing Ca^{2+} absorption by the intestines.







**note: Thyrocalcitonin works in balance with Parathyroid Hormone (PTH) produced by the Parathyroid.*

Learning Aim: Explain the effects thyrocalcitonin has on the body